Vaishnov Nikhil Kadiyala

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EDUCATION

The University of Texas at Austin, Austin, TX **Anticipated May 2029**

Bachelor of Science in Informatics, Human-Centered Data Science & Cybersecurity

GPA: 3.60

Relevant Coursework: Human-Centered Data Science, Elements of Computers and Programming

EXPERIENCE

SUSTAINABLE BUILDING INITIATIVE

Tech Director **Aug 2026 - Present**

- Lead a 10 to 15 person tech team building the org website, approving all work before it merges and running recruiting, interviews, and onboarding
- Scoped roughly $15K in lab GPU funding, pricing an enterprise Dell build against a workstation tier and pitching the faculty principal investigator a more cost-efficient path through UT Austin's TACC allocation

Tech Lead, Drone Road-Inspection Research **May 2026 - Present**

- Lead tech for a ~10 person research team detecting road cracks from drone imagery and building a georeferenced road map
- Optimized a remote PyTorch training pipeline on an NVIDIA RTX 3080, resolving GPU memory and data-loading bottlenecks to accelerate model training by ~25x
- Trained a YOLO26-Large crack detector on a 5,406-image dataset at 1280 px, improving mAP50 from the team baseline of 0.57 to 0.678 at 0.635 recall
- Co-author on the team's road-crack detection manuscript, in preparation

PERSPECT **Jan 2026 - Present**

Founder

- Designed and shipped a solo full-stack AI product where expert AI personas debate a topic, map their disagreements, and output a cited briefing; top 20 at the Hooked on Claude 2026 Hackathon
- Built a Bun + Express backend streaming a multi-persona pipeline over Server-Sent Events on Gemini 2.5 with Google Search grounding, plus a React + Tailwind frontend, deployed on Vercel + Render with CI

BIZFIRST AI **Jun 2026 - Aug 2026**

Software Engineering Intern

- Shipped BizFirst's first self-hosted object storage tier on MinIO in Docker with tiered quotas, 365-day lifecycle expiration, and restic snapshots to Backblaze B2 for offsite backup
- Built three MCP tools that let BizFirst's multi-agent orchestrator create and update forms from natural language, replacing hand-written JSON schemas and direct REST calls

ICELIS GLOBAL **Jun 2026 - Jul 2026**

ESG Consulting Intern

- Delivered ~90 slides of client training on IFRS S1/S2, GRI Standards, and GCC regulations, including a materiality-assessment playbook and an 8-week engagement plan

PROJECTS

NETSENTINEL | *Python, FastAPI, XGBoost, React, TypeScript, Docker, Kubernetes*

- Built a real-time intrusion detection system with an XGBoost multi-class classifier on NSL-KDD (125K+ samples, 77% accuracy, 96% DoS precision)

GRAVITY-ZERO | *Python, FastAPI, React, LM Studio*

- Built a fully local AI IDE with a Planner, Coder, and Reviewer agent triad on a quantized Qwen 2.5 Coder 7B model
- Ran the whole pipeline offline through LM Studio with a FastAPI backend and a React frontend, keeping source and prompts on device with no cloud API calls

LEADERSHIP & COMMUNITY INVOLVEMENT

TEXAS NOVA CONSULTING, Austin, TX **Sep 2026 - Present**

Tech Analyst

- Selected via a competitive application and panel case interview for UT's AI and technology consulting organization
- Apply the consulting curriculum to hands-on client engagement work as the technical analyst, scoping the problem and building the deliverables with the team

HC4A: AI WITH ALL HACKATHON, Austin, TX **Jul 2026 - Present**

Tech Lead

- Co-lead the technology and student council side of a ~600-participant youth hackathon for grades 8 to 12, designing the participant app and a privacy-aware UID scheme under COPPA and FERPA

HONORS AND AWARDS

- Top 20, Hooked on Claude 2026 Hackathon | 5th Place, UT MLDS DataHack 2026 | National Cyber Scholar 2024 | 6x CyberPatriot State Semifinalist | SANS Foundations, GIAC certified

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS, POSIX Shell, Java, C#/.NET, Lua

Technologies: React, Next.js, FastAPI, Node.js, PostgreSQL, Supabase, Linux, Docker, Kubernetes, PyTorch, CUDA, Ultralytics YOLO, OpenCV, XGBoost, scikit-learn, LLM APIs (Gemini, Claude), MCP